Total No. of Questions : 8]

SEAT No. :

P2905

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[6004]-804

B.E. (Computer Engineering)

EMBEDDED AND REAL TIME OPERATING SYSTEMS

(2015 Pattern) (Semester - II) (Elective - III) (410252 (C))

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Solve questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicates full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) Explain block diagram of Embedded Systems. **[6]**

b) Explain in detail ARM Processor Architecture with core architectural block diagram. **[6]**

c) Write short note on ISA and PCI. **[8]**

OR

Q2) a) Explain Hardware Components of an Embedded System. **[6]**

b) How does ARM micro-controller differ from SHARC processor? Justify your answer. **[6]**

c) Describe and Compare RS232C and SDIO Devices. **[8]**

Q3) a) How to represent precedence constraint and data dependency among real time tasks? Explain with diagram. **[8]**

b) What is RTOS? Explain hard versus soft real time systems and their timing constraints. **[8]**

OR

Q4) a) Explain least slack time first scheduling and latest release time scheduling in real time systems. **[8]**

b) Differentiate between fixed priority and dynamic priority scheduling algorithms in real time systems. Give one example of each. **[8]**

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Q5) a) What is priority inversion problem in real time systems? How this problem can be solved? Give details. **[8]**

b) Explain message queues with suitable diagram. **[8]**

OR

Q6) a) What is Semaphore? How does it help in resource sharing in RTOS kernel? **[8]**

b) Explain with example interrupts enabling and disabling in embedded system. **[8]**

Q7) a) Explain with example Validation and Debugging in an embedded system. **[6]**

b) What are issue in resource reservation? Explain resource reservation protocol with diagram. **[6]**

c) Write short notes on MicroC/OS-II, Windows CE? **[6]**

OR

Q8) a) Explain priority based service disciplines for switched networks. **[6]**

b) Explain Medium access control (MAC) protocol for broadcast networks. **[6]**

c) Write Short Notes on RT Linux, VxWorks? **[6]**